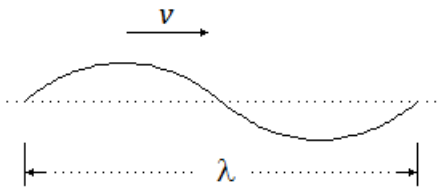
WAVES

Preparation:

Waves occur in many forms ranging from mechanical waves such as earthquakes, ocean waves, sound waves, vibrations in materials and so on, to electromagnetic and perhaps gravitational waves. All waves have some sort of pattern that repeats continuously and one of the simplest forms is the sine function shown in figure 1 where λ is the wavelength and v is the velocity. The rate at which the waves pass a given point is the frequency f and equation 1 describes the simple relationship that applies to **all** waves.

Fig. 1



$$v = \lambda f \qquad \text{Eq. 1}$$

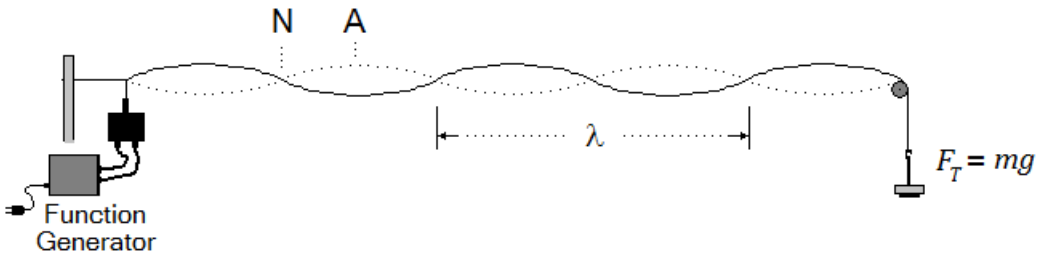
WAVES IN STRING:

Many different variables affect the various types of waves. For waves in string the tension in the string F_T and the mass per unit length of the string μ both affect their velocity along the string. The relationship is described in equation 2.

$$v = \sqrt{\frac{F_T}{\mu}} \qquad \text{Eq. 2}$$

Standing waves can be created by reflecting waves off the fixed ends of a string as in figure 2. If the space provided is just right for the wavelength, the crests of the waves heading toward the ends and the crests of the waves reflected back from the ends add together at specific points along the wire (**antinodes**) and cancel each other at specific points along the wire (**nodes**). This condition is called **resonance**.

Fig 2



Procedure:

1. Obtain a piece of string long enough to reach across the lab table and hang over the side about 20 or 30cm. Measure its stretched out length (with **no** knots!). Measure its mass, and calculate μ = total mass / total length.

2. Assemble the apparatus shown in figure 2.

3. Turn on the function generator and turn the frequency down to zero. Slowly increase the frequency until a large *single* antinode (A) occurs. This is called the fundamental mode. Note this frequency and then slowly continue increasing the frequency until the next pattern appears composed of *two* antinodes with a single node (N) in between them.

   a) How did the resonant frequencies increase? Did the frequency double, or did it just increase by some particular amount? Can you be sure with the information you have so far?

4. Increase the frequency to the third resonance (three antinodes).

   a) Can you determine how the resonant frequencies are increasing now?

   b) Noting the frequencies of these first three resonances, estimate the frequencies where the fourth, fifth, and sixth resonances should occur. Slowly increase the frequency and verify these frequencies.

5. Set the frequency to the fifth resonance. Slowly pull down on the weight hanger manually to increase the tension.

   a) What happens to the antinodes as the tension slowly increases?

   b) Increase the tension until resonance returns. The **number** of antinodes *increased*, *decreased*, or *stayed the same*?

   c) The frequency is the same, but the velocity has changed per equation 2 above. What should this do to the wavelength of the waves?

   d) Does your prediction in part c concur with your experimental observations? If not, retrace your reasoning through equations 1 and 2.

6. Let go of the hanger and allow the string to return to the fifth resonance. Predict what will happen if you slowly lift up on the hanger to decrease the tension.

a) Does your prediction concur with your experimental observations? If not, retrace your reasoning through equations 1 and 2 once again.

7. Once again, let go of the hanger and allow the string to return to the fifth resonance. Now sketch this standing wave pattern in your report, and label all the nodes and antinodes. Determine λ directly by measuring all the distances between nodes and finding the average value. Note carefully from figure 2 that this value represents $\lambda/2$. Use equation 1 to calculate the wave velocity in the string v_{eq1} using this average λ and the frequency of the function generator.

8. Use equation 2 to calculate the wave velocity in the string v_{eq2} based on μ and F_T.

   a) Calculate the % difference between v_{eq1} and v_{eq2} in string.

   b) At this speed, how long does it take a wave to travel from one end of the string to the other and back again? How many times does it do this each second?

WAVES IN AIR:

Standing waves of sound can be created by reflecting them off the bottom of a long tube as in figure 3. For waves in the air, the temperature affects the velocity of sound. The relationship is described in equation 3, where 331m/s is the speed of sound at a temperature of 273K (0°C).

$$v = 331\frac{\text{m}}{\text{s}}\sqrt{\frac{T_{air}}{273\text{K}}} \qquad \text{Eq. 3}$$

Procedure:

1. Set up the long tube and bring the reservoir to the very top. Fill it with water until the level in the tube is at about 10cm from the top. Lower the reservoir to lower the water level in the tube, allowing the length of the chamber to increase.

2. Lower the water level while ringing a tuning fork at the top of the tube. Find the locations on the scale of L_1 and L_2 where the **loud resonances** clearly occur. Try to reach L_3 but keep an eye on the water level in the reservoir to prevent spillage. Remove some water if necessary to reach L_3.

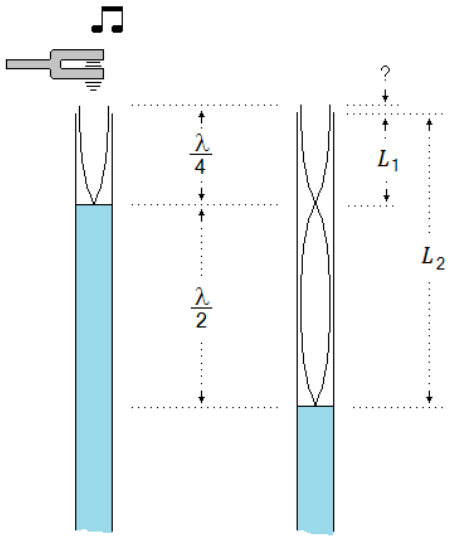


Fig. 3

3. Determine the experimental λ by measuring the distances between nodes and finding the average value. Note carefully from figure 3 that this value represents λ/2. Do **not** assume L_1 = λ/4. Use equation 1 to calculate the wave velocity in air v_{eq1}, using this average λ and the frequency stamped on the fork.

4. Measure the air temperature near the tube and use equation 3 to calculate the wave velocity in air v_{eq3} based on T_{air}.

   a) Calculate the % difference between v_{eq1} and v_{eq3} in air.

   b) At this speed, how long does it take a wave to travel down to the third resonance depth and back again? How many times does it do this each second?

   c) Considering the two examples dealt with here, strings and tubes, list some situations where you might *want* resonance.

   d) Consider some of the other types of waves mentioned in the first paragraph above, and list some situations where you might want to *avoid* resonance.